

Probability Without a Joint World

The survey's open problem concerns a very specific object — a **σ -additive, σ -essentially contextual state on a concrete, σ -complete orthomodular lattice** — but its real subject is much older and much larger. It asks what becomes of probability when the “events” to which probabilities are assigned no longer form one distributive family of jointly testable alternatives. In the classical setting, that failure never even appears: one begins with a field or σ -field of events and a probability measure on it. In the orthomodular setting, by contrast, one begins with overlapping **contexts** of jointly performable observations and then asks whether those local pieces support a single global probability theory. The survey takes that question to be genuinely open in one sharply delimited cell, and what follows is an orientation to why that remaining cell matters.

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The question behind the question

Kolmogorov's own advice is the cleanest point of departure: when formulating a probabilistic problem, one must say **for which events probabilities are assumed to exist**. In his *Foundations*, Axioms I–V define probability on a field of sets with finite additivity, while the continuity axiom for infinite fields is introduced separately as Axiom VI and stated to be independent of I–V for infinite event-fields. He also remarks that the enlarged Borel event-families are often “ideal events” to which nothing directly corresponds in the outside world. In other words, the move from finite coherence to full σ -additivity is not forced by elementary consistency alone; it is an additional structural commitment about the range of events one is prepared to countenance. ¹

That is exactly where de Finetti enters. The later literature on his view is unambiguous: he did **not** think countable additivity follows from coherence as such, and he rejected the idea that Dutch-book style consistency by itself compels σ -additivity. More recent work on countably many bets sharpens the same point: finite betting coherence supports finite additivity automatically, but extra assumptions are needed before one can recover countable additivity from betting considerations. The deepest lesson of the classical case, then, is not that σ -additivity is optional in practice, but that it is **not forced by finitary rationality alone**. ²

The survey's open problem asks for the non-distributive analogue of that old issue. In a Boolean algebra of events, the global sample-space picture is built into the logic of events themselves. In an orthomodular lattice, finite pieces can still look perfectly classical inside each compatible block, while the whole need not admit a single globally joint description. The question becomes: **when do locally coherent probability assignments force a genuinely σ -additive global probability, and when does that implication fail once joint observability fails?** The survey's “ σ -essential” refinement matters because Wright's pentagon already shows that finite contextuality exists; what is open is whether contextuality can exist only at the countable level, so that every finite piece remains classically interpretable while the countable whole does not.

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The two dividing lines

Two distinctions organize the entire landscape. The first is **distributivity**, which is the lattice-theoretic face of **commutativity**: inside a commuting family of observables one has a Boolean algebra, but once incompatible observables are glued together the event-structure becomes non-distributive. The second is **concreteness**, meaning not “having geometric points” in the sense of rays or atoms, but having enough **two-valued valuations** — dispersion-free states — to determine the order. In classical Stone duality these notions of point coincide nicely; in quantum logic they come apart. ³

The Hilbert-space case is the clearest warning against conflation. The lattice of closed subspaces, or equivalently the projection lattice, is rich in geometric points: it has abundant rays, and Gleason’s theorem shows that σ -additive probability measures on this lattice are exactly the trace-form states in dimensions at least three. But Kochen–Specker blocks a different kind of point: a globally consistent, noncontextual two-valued valuation on all projections in dimension at least three. So the Hilbert lattice is geometrically point-rich and valuation-poor at the same time. That is why “point” must be handled with care here. ⁴

The decisive 2x2 is therefore not “classical versus quantum” in one stroke, but **distributive versus non-distributive** crossed with **concrete versus non-concrete**:

	Distributive	Non-distributive
Concrete	Classical probability. Boolean algebras or σ-algebras with Stone-type point semantics: operationally, all relevant events can be jointly embedded in one sample space. ⁵	The open cell. Concrete non-Boolean logics have enough two-valued states to separate order, yet incompatible observations prevent a single distributive event algebra. The product example $\prod_n MO_2$ lies here, but the survey shows it is segregated and does not yield σ-essential contextuality. Operationally, there are global valuation-points, but infinitary contextuality still fails. filecite turnOfile0
Non-concrete	Point-free but still distributive. Measure algebras and localic/locale-based measure theory remove ordinary points without leaving the distributive world. Operationally, one may have an excellent σ-additive measure theory, but it lives on frames/locales rather than on valuation-rich set models. ⁶	Hilbert-space quantum logic. The lattice $L(H)$ is non-distributive, Gleason supplies σ-additive states, and Kochen–Specker forbids a global noncontextual valuation in dimension at least three. Operationally, probabilities exist, but they are not mixtures over global definite-value assignments. ⁴

Each cell has a distinct operational meaning. The upper-left cell is the ordinary world of one joint observation-space. The lower-right cell is the standard quantum world: native noncommutativity, real σ -additive states, but no global value assignment of the Kochen–Specker kind. The lower-left cell shows that one can be point-free without being non-distributive; localic measure theory is already a mature example of that. The survey’s open problem lives in the upper-right cell: **non-distributive and yet concrete**. That is the

one place where enough valuation-points survive to make a hidden-variable reading tempting, but where distributivity still fails badly enough that a global sample space may not exist. ⁷

This is why the user’s requested pairing of the two axes is so important. Non-distributivity by itself is not enough, because $L(H)$ already occupies the non-concrete pole. Concreteness by itself is not enough, because Boolean algebras are concrete in the boring classical way. The unresolved question is whether there is a **concrete** non-Boolean σ -complete orthomodular lattice whose σ -additive probability theory is genuinely contextual in a way that no finite substructure can witness. That is an extremely special target, but it is also the only remaining target that could support a truly non-Boolean, point-free-yet- σ -additive probability theory. [filecite turn0file0](#)

The Loomis–Sikorski hinge

In the Boolean case, the bridge from event-logic to ordinary measure theory is so familiar that it becomes invisible. Stone’s representation turns a Boolean algebra into the clopen algebra of a compact totally disconnected space whose points are ultrafilters, equivalently two-valued homomorphisms. The measure-theoretic story then runs through representation and extension: once the event algebra is represented by sets of points, σ -additivity becomes ordinary measure theory on that space or on an equivalent quotient presentation. This is why classical probability can afford to look effortlessly both algebraic and spatial. ⁸

The survey’s central insight is that, in the orthomodular setting, the whole problem turns on whether anything like that Boolean engine survives. McDonald and Bimbó do provide a genuine topological duality for orthomodular lattices, representing them as $\perp$ -stable clopen sets of suitable orthomodular spaces. But this duality is **finitary**: it preserves the lattice operations that belong to the orthomodular algebraic structure, not the countable joins that σ -additivity needs for a full measure-theoretic descent. The survey’s terminology of “extension” and “descent” is built precisely around that gap. ⁹

The first hard theorem at this hinge is negative and simple to state. A **faithful** set representation of an orthomodular lattice — one preserving meets, joins, and orthocomplements as literal intersection, union, and complement — would place the lattice inside an honest powerset algebra. But powersets are distributive. Hence a faithful set representation of this kind forces the lattice to be Boolean. This is not just one failed construction strategy; it is a structural closure result. The straightforward “let the events really be sets of points” move is closed off for every genuinely non-Boolean orthomodular lattice.

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That leaves a weaker possibility: **concrete logic** in the sense used by Gudder and the survey. Here the lattice or poset is representable by sets only in a reduced sense, preserving complement and orthogonal sums rather than all finite lattice operations, or equivalently carrying an order-determining family of two-valued states. This is the non-faithful, non-Boolean style of concreteness that survives the distributivity obstruction. It is exactly the regime in which the open problem lives. A positive answer would have to work here, not by recovering full set-theoretic faithfulness, but by extracting enough σ -additive structure from these weaker “points” to support a genuinely relational probability theory. [filecite turn0file0](#)

There is a second theorem that tightens the noose further. Pták and Pulmannová proved that an orthomodular lattice with a **unital set of subadditive probability measures** must already be distributive, hence Boolean. So even a finitary strengthening of probabilistic behavior pushes the structure back across

the distributive boundary. The survey uses this theorem as part of its explanation of why the open problem is delicate: if a witness exists, it cannot be found by merely strengthening the Boolean-looking side of the theory. It must live exactly in the narrow space left over after the faithful spatial route and the subadditive route have both collapsed to Booleanity. ¹⁰

Why the remaining boundary is so hard to move

One reason the problem is compelling is that several quite different research programs all converge on the **same obstruction**.

The first route is lattice duality itself. Existing orthomodular dualities are rich enough to recover the underlying lattice from a topological or relational space, but they are not σ -dualities of the Stone–Loomis–Sikorski kind. McDonald–Bimbó’s duality is explicitly a dual equivalence for orthomodular lattices and certain orthomodular spaces, yet it does not supply the countable-join-preserving measure apparatus one would need to push σ -additive states down to measures on ordinary points. Meanwhile the MacNeille-completion route, which in distributive settings often helps absorb missing suprema, breaks: Harding showed that an orthomodular lattice may have a MacNeille completion that is **not** orthomodular. So one cannot simply complete the lattice and expect the σ -story to repair itself upstairs. ¹¹

A related obstruction appears from the opposite side, in noncommutative spectrum theory. Reyes, and later Ben-Zvi–Ma–Reyes, showed that attempts to extend commutative spectrum functors to genuinely noncommutative rings or C^* -algebras run into Kochen–Specker type emptiness results: any such contravariant spectrum functor assigning ordinary sets, spaces, or locales must send full matrix algebras $M_n(R)$ for $n \geq 3$ to the empty object. The details belong to algebra rather than lattice theory, but the moral is the same: once one insists on a spectrum of “points” that behaves classically on the commutative part, native noncommutativity tends to destroy that spectrum rather than extend it. ¹²

The second route is topos theory and Bohrification. This is the most sophisticated existing machinery for making quantum probability look point-free and geometric, and it matters enormously because it shows how much can already be done. Heunen, Landsman, and Spitters replace a noncommutative algebra by an **internal commutative** algebra inside a topos; its spectrum is then an internal locale, and the logic of propositions becomes a Heyting algebra, hence distributive. Döring’s spectral presheaf plays an analogous role in the contravariant approach: every quantum state defines a measure on clopen subobjects of the presheaf, and abstract such measures recover states. Wolters’ comparison makes the common theme explicit: both major topos approaches are built over the poset of **commutative contexts**. ¹³

That achievement is profound, but it is not the same as a native σ -additive probability theory on a non-distributive logic. Bohrification recovers distributivity internally; it does not preserve non-distributivity as the home of the valuation. Döring’s measures live on clopen subobjects of the spectral presheaf rather than on the original orthomodular lattice as such, and the survey is right to treat this as a sidestep rather than a solution. The topos program shows that point-free, geometric, and even σ -like behavior are available **after moving to a distributive object of contexts**. The open problem asks whether one can stay on the non-distributive side and still obtain genuinely σ -additive probability. ¹⁴

The third route comes from imprecise probability and measure extension. Derr and Williamson’s framework is especially important because it translates “quantum” extension questions into the language of

probabilities on pre-Dynkin and Dynkin systems — systems of events on which probability is only partially or contextually defined. They prove that, under extendability conditions, such systems embed into algebras of sets, and they identify a lattice duality relating systems of precision to credal sets. In their countably additive appendix, they then assume a **Polish** underlying space, a Dynkin system inside its Borel σ -algebra, and inner regularity on the σ -blocks; under those hypotheses, Maharam-style extension machinery applies, the measure extends to the ambient Borel σ -algebra, and the countably additive problem is settled negatively for that representable regime. ¹⁵

This is the crucial topological hinge, and it has to be described carefully. What is ruled out is **not** “everything non-standard-Borel,” and the survey is right to warn against that shortcut. Nor is the residue merely “non-spatial” in some loose categorical sense. The precise statement is that the remaining frontier lies beyond the cases that admit a suitable **Polish** realization, or more generally the **Hausdorff plus inner-regular** setting on which the extension theorem runs. So if a witness exists, it must fail to admit that kind of regular point realization. The right inference is not simply “therefore non-Polish,” still less “therefore exotic but otherwise ordinary.” The honest residue is more severe: a witness would have to come with a **descriptively wild** state space, wild enough to evade the regularity assumptions that make Maharam-type extension possible. That is why the survey says the remaining question has genuinely become one in descriptive set theory. ¹⁶

What is really at stake

A **positive** answer would do more than fill a gap in the literature. It would exhibit a genuinely **relational, point-free, σ -additive probability theory on a non-distributive logic**. There are already point-free σ -additive measures on distributive frames and locales — this is exactly what localic measure theory studies — but that success always takes place on a distributive object. A witness in the survey’s open cell would say something structurally new: one can retain countable additivity even when the logic of events itself remains natively non-Boolean and no faithful set-theoretic realization is available. In that sense, it would be a non-Boolean analogue of localic measure theory rather than a variant of ordinary sample-space probability. ¹⁷

A **negative** answer would also be conceptually strong. It would amount to a σ -essential impossibility theorem: whenever a concrete σ -complete non-Boolean logic carries σ -additive probabilities, any genuinely contextual behavior would either already appear finitely or else vanish once one asks for countable additivity. The philosophical point would not be that non-Boolean logics are unreal or useless. It would be narrower and more instructive: the step from finitely coherent contextual data to a determinate σ -additive global probability is not fixed by operational structure alone. In that sense, the relational content would be **operationally underdetermined** unless one adds a completion principle from outside the finite observational net. ¹⁸

This is where van Fraassen’s distinction between **empirical adequacy** and further realist completion is illuminating, though it should not be overextended. Van Fraassen’s idea is that a theory can succeed by correctly accommodating observable phenomena without thereby committing one to a unique underlying ontology. The survey’s open problem is not a direct application of constructive empiricism, but it sits near the same fault line. A positive answer would show that one can have a countably additive probability theory whose content remains genuinely relational, not merely shorthand for a hidden space of definite states. A negative answer would instead suggest that beyond finite operational structure, σ -additive completion requires precisely the kind of extra representational layer that an empiricist would treat cautiously. The

parallel is suggestive because the question is not whether local observations are meaningful — they plainly are — but whether they determine a unique σ -additive global “world behind them.” ¹⁹

On present evidence, the **negative** outcome does look somewhat more plausible. Several routes that might have produced a positive theory already collapse back to the distributive world: faithful point realizations force Booleanness; unital families of subadditive states force Booleanness; MacNeille completion does not rescue σ -structure in the orthomodular category; topos methods recover distributivity internally rather than preserving native non-distributivity; and the representable, regular topological case is already closed by the Derr–Williamson/Maharam machinery. None of that is a proof against the open cell. But it does explain why the remaining frontier is so narrow and why, if a witness exists, it is likely to be conceptually stranger than the familiar examples from Hilbert space, Bell scenarios, or finite contextual graphs. ²⁰

The honest state of play

What is **proved** is already substantial. Classical probability depends on specifying an event-domain and adding σ -structure deliberately. In the orthomodular world, noncommutativity produces non-distributive event-logics; the Hilbert lattice supplies σ -additive states via Gleason but lacks global two-valued valuations by Kochen–Specker; faithful set representations of non-Boolean orthomodular lattices are impossible; and the known extension machinery rules out the remaining open cell whenever the concrete logic admits a sufficiently regular point realization of the Polish or Hausdorff-plus-inner-regular sort. Existing point-free frameworks either stay distributive from the start or recover distributivity by moving to contexts. ²¹

What remains **open**, on the assumptions fixed by the prompt and the survey, is one sharply bounded possibility: a concrete, σ -complete, non-Boolean orthomodular lattice carrying a σ -additive state that is contextual globally but noncontextual on every finite sub-OML. That is not merely one more search problem among many. It is the cleanest place where the two oldest foundational pressures on probability — the passage from finite coherence to σ -additivity, and the passage from compatible local tests to a single global event-space — are forced to meet without being allowed to collapse into one another. Whether this last cell is inhabited or empty is therefore a test of how far probability can remain relational before it must choose between distributive space and underdetermination. ²²

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